

Development Process Report

Smart School Attendance & Parent Notification System

Node.js | Express.js | MySQL | EJS | Bootstrap

Prepared for project submission

Report Contents

- Project title
- App idea
- Purpose of the project
- AI tools used
- Initial prompt used
- Prompt iterations used during development
- What worked well
- What did not work
- Bugs or challenges faced
- How the issues were solved
- Features added or improved
- Final result of the project
- Whether the project solves a real-world problem

1. Project Title

Smart School Attendance & Parent Notification System

2. App Idea

The project is a database-based school attendance management portal. It allows school administrators to manage teachers, parents, students, and classes. Teachers can mark attendance for students in their assigned classes. When a student is marked absent, the system automatically creates an in-app notification for the linked parent.

The application has three main user portals: Admin Portal, Teacher Portal, and Parent Portal. Each user logs in using an email and password, and the system redirects the user to the correct dashboard according to their role.

3. Purpose of the Project

The purpose of this project is to solve a common school communication problem. Many schools take attendance manually, and parents may not be informed immediately when their child is absent. This system makes the attendance process faster, more organized, and more transparent for parents.

The project also demonstrates how a real-world web application can be developed using Node.js, Express.js, MySQL, EJS templates, Bootstrap, sessions, password hashing, and role-based access control.

4. Technology Stack

Technology	Purpose
Node.js	Backend JavaScript runtime
Express.js	Web application framework and route handling
MySQL	Database for users, students, classes, attendance, and notifications
EJS	Server-side template rendering
Bootstrap	Responsive user interface design
express-session	Login session management
bcrypt	Secure password hashing
mysql2	MySQL database connection
dotenv	Environment variable management

5. AI Tools Used

The main AI tool used during the development process was ChatGPT. It was used to plan the project structure, generate beginner-friendly Node.js code, create MySQL database schema, prepare authentication logic, build role-based dashboards, and troubleshoot deployment-related questions.

AI was used as a development assistant, but the project idea, requirements, testing, deployment decisions, and final review were guided by the developer.

6. Initial Prompt Used

The initial prompt asked ChatGPT to create a database-based School Attendance Management and Parent Notification Portal using Node.js, Express.js, MySQL, EJS, Bootstrap, JavaScript, express-session, bcrypt, mysql2, and dotenv.

The prompt clearly required that all main data should be stored in MySQL and that localStorage should not be used for users, students, classes, attendance records, or notifications. It also described the three roles: Admin, Teacher, and Parent.

The initial prompt also listed the main features, database tables, relationships, project folder structure, UI requirements, seed users, attendance logic, and security requirements.

7. Prompt Iterations Used During Development

Several prompt iterations were used to improve and complete the project. These iterations helped divide the work into smaller modules and solve specific problems step by step.

Iteration	Prompt Focus	Result
1	Generate complete project structure	Created organized folders for routes, controllers, middleware, views, config, and public assets.
2	Create database schema and seed data	Created MySQL tables for users, classes, students, parent_student, attendances, and notifications.
3	Build login system	Added session-based login, bcrypt password checking, logout, and role-based redirects.
4	Develop admin module	Added dashboards and CRUD management for teachers, parents, students, classes, and reports.
5	Develop teacher attendance module	Allowed teachers to see assigned classes, mark attendance, update existing attendance, and prevent duplicate attendance records.
6	Develop parent module	Allowed parents to view linked children, attendance history, notifications, unread count, and mark notifications as read.
7	Deployment guidance	Prepared steps for GitHub, environment variables, hosting, and Rocky Linux server deployment.
8	UI cleanup	Removed demo username and password details from the login page for a cleaner public interface.

8. What Worked Well

- The project requirements were clear, which made it easier to divide the system into Admin, Teacher, and Parent modules.
- The MySQL database structure worked well because each major feature had its own table and relationships were easy to understand.
- Using EJS made it simple to render dynamic pages such as dashboards, forms, tables, and reports.
- Bootstrap helped create a clean and responsive dashboard interface quickly.
- Role-based middleware worked well for protecting routes and preventing users from accessing another role's pages.
- Attendance update logic worked well because duplicate attendance records were prevented using a unique rule for student, class, and date.

9. What Did Not Work

- The application could not be hosted on GitHub Pages because GitHub Pages only supports static websites and this project needs a running Node.js backend server.
- Some free hosting platforms asked for a credit card, so they were not suitable for no-card deployment.
- A normal .env file could not be pushed to a public repository because it contains private database credentials and session secrets.
- The default express-session memory store is not suitable for production hosting, especially on serverless platforms.

10. Bugs or Challenges Faced

Challenge	Details
Database security	The application needed to use database credentials without exposing them in a public GitHub repository.
Hosting selection	GitHub Pages could not run Express.js and some free platforms required a card.
Attendance duplication	The system needed to avoid creating multiple attendance records for the same student, class, and date.
Role security	Teachers and parents needed to be restricted from accessing data that did not belong to them.
Notification logic	A notification had to be created only when a student was marked absent and linked parents existed.
Login page privacy	Demo login credentials were displayed below the login form and needed to be removed before public use.

11. How the Issues Were Solved

- The .env file was added to .gitignore so private credentials are not committed to GitHub.
- An .env.example file was used to show required configuration keys without exposing real secrets.
- For deployment, the recommendation was changed to a self-hosted Rocky Linux 9 server when free hosting platforms required a card.
- Nginx was planned as a reverse proxy, while Node.js runs internally on localhost:3000 using PM2.
- Attendance records were saved or updated instead of blindly inserted, preventing duplicate records.
- Middleware was used to protect routes and check the logged-in user role before allowing access.
- Teacher queries were restricted to assigned classes only, and parent queries were restricted to linked children only.
- The login page was cleaned by removing the demo credentials block from views/login.ejs.

12. Features Added or Improved

- Admin dashboard with summary cards for teachers, students, parents, classes, and attendance records.
- Admin CRUD management for teachers, parents, students, and classes.
- Teacher dashboard showing only assigned classes.
- Attendance marking form with Present and Absent status options.
- Automatic absent notification creation for linked parents.
- Parent dashboard with linked children, attendance history, notification list, unread count, and mark-as-read option.
- Attendance report filters by class, student, date, and status.
- Bootstrap UI with sidebars, tables, forms, and colored badges for Present and Absent statuses.
- Secure login using hashed passwords and session-based authentication.

13. Final Result of the Project

The final result is a working database-based school attendance and parent notification web application. Admin users can manage the school data, teachers can mark and update attendance, and parents can view their child's attendance history and receive in-app absence notifications.

The application uses MySQL as the main database and does not depend on localStorage for core data. It includes role-based dashboards, protected routes, secure password hashing, session login, and a clean Bootstrap interface.

14. Does the Project Solve a Real-World Problem?

Yes. The project solves a real-world school communication problem. In many schools, attendance is still managed manually and parents may not know immediately when their child is absent. This system improves communication by automatically notifying parents when a teacher marks their child absent.

The project can be useful for small schools, tuition centers, academies, and training institutes. It can also be expanded in the future with SMS, WhatsApp, email notifications, homework management, assignments, quiz marks, exam results, timetable, fee management, and PDF reports.

15. Conclusion

The Smart School Attendance & Parent Notification System is a practical full-stack web application that demonstrates database design, authentication, authorization, CRUD operations, attendance management, and notification handling. It is beginner-friendly but still follows important real-world concepts such as role-based access, secure passwords, protected routes, and environment variable security.

Overall, the project successfully meets its goal of making school attendance management easier and helping parents receive timely updates about student absences.